In[∘]:=
```
L = 7 / 10 ;
S1 = 3 / 10 ;
S2 = 4 / 10 ;
```

If $m_1, m_2 > 0$ and $m_1 > m_2$, then $0 < \sigma_1 < 1.75$

In[∘]:=
```
σ1 = 15 / 10  ; (*σ1=1.5*)
0 < σ < 175 / 100 /. σ → σ1
```

Out[∘]=

```
True
```

Parameter definitions

In[∘]:=

$$\Delta 1 = \frac{\left(\frac{(J^2-L^2-S1^2-S2^2)}{2} - \frac{SeffdL}{\sigma 2}\right)}{\sigma 1 - \sigma 2};$$

$$\Delta 2 = \frac{\left(\frac{(J^2-L^2-S1^2-S2^2)}{2} - \frac{SeffdL}{\sigma 1}\right)}{\sigma 1 - \sigma 2};$$

$$\sigma 2 = 1 + \frac{1}{(-1 + \sigma 1)};$$

```
(*σ1=(1+ m2/m1)   σ2=(1+ m1/m2)*)
(* Not this σ1=(1+3/4 m2/m1)   σ2=(1+3/4 m1/m2)*)
```

negación

Cubic function roots

In[ ]:=
```
cubic = Expand[ L^2 S1^2 S2^2 - 2 σ1 σ2 (σ1 - σ2) f (f - Δ1) (f - Δ2) -
                expande factores
    L^2 (σ1 - σ2)^2 f^2 - S2^2 σ2^2 (f - Δ1)^2 - S1^2 σ1^2 (f - Δ2)^2 ]
 (*The cubic polynomial of Eq. 3.1 of Cho-Lee*);
a3 = Coefficient[cubic, f^3];
     coefficiente
a2 = Coefficient[cubic, f^2];
     coefficiente
a1 = Coefficient[cubic, f];
     coefficiente
a0 = Expand[cubic - (a3 f^3 + a2 f^2 + a1 f)];
     expande factores
p = (3 a1 a3 - a2^2)/(3 a3^2);
q = (2 a2^3 - 9 a1 a2 a3 + 27 a0 a3^2)/(27 a3^3);
roots = Table[f_k → -a2/(3 a3) + 2 Sqrt[-p/3] Cos[1/3 ArcCos[3 q/(2 p) Sqrt[-3/p]] + 2 π k/3], {k, 1, 3}];
        tabla                  coseno arco coseno
f1 = roots[[1, 2]];
f2 = roots[[2, 2]];
f3 = roots[[3, 2]];
```

In[ ]:=

Flow amounts definitions

In[ ]:=
```
(*The following definitions are in Cho-Lee*)
```

$\Delta\lambda_{S_{\text{eff}}.L}$

In[ ]:=
```
A = 2 σ1 σ2 (σ2 - σ1);
β = Sqrt[(f2 - f1)/(f3 - f1)] (*elliptic modulus "k"*);
Δλ1 = 4/Sqrt[A (f3 - f1)] EllipticK[β];
                          K elíptica
```

$\Delta\lambda_{J^2}$

In[ ]:=
$$B1 = \frac{1}{2}\left(\left(SeffdL + L^2(\sigma1+\sigma2)\right)(J+L) + L\left(\sigma1\, S1^2 + \sigma2\, S2^2 + (\sigma1+\sigma2)(\Delta2\,\sigma1 - \Delta1\,\sigma2)\right)\right);$$
$$B2 = \frac{1}{2}\left(\left(SeffdL + L^2(\sigma1+\sigma2)\right)(J-L) - L\left(\sigma1\, S1^2 + \sigma2\, S2^2 + (\sigma1+\sigma2)(\Delta2\,\sigma1 - \Delta1\,\sigma2)\right)\right);$$
$$D1 = L(L+J) + \Delta2\,\sigma1 - \Delta1\,\sigma2;$$
$$D2 = L(L-J) + \Delta2\,\sigma1 - \Delta1\,\sigma2;$$
$$\alpha1 = \frac{(\sigma1-\sigma2)(f2-f1)}{D1 - f1(\sigma1-\sigma2)};$$
$$\alpha2 = \frac{(\sigma1-\sigma2)(f2-f1)}{D2 - f1(\sigma1-\sigma2)};$$
$$\Delta\lambda2 = -\frac{1}{2J}\left(\frac{4}{\sqrt{A(f3-f1)}}\right)\left(\frac{B1\ \text{EllipticPi}[\alpha1,\beta^2]}{D1 - f1(\sigma1-\sigma2)} + \frac{B2\ \text{EllipticPi}[\alpha2,\beta^2]}{D2 - f1(\sigma1-\sigma2)}\right);$$

$\Delta\lambda_{L^2}$

In[ ]:=
$$\Delta\lambda3 = -\frac{1}{2L}\left(\left(\frac{4}{\sqrt{A(f3-f1)}}\right)\left(\frac{B1\ \text{EllipticPi}[\alpha1,\beta^2]}{D1 - f1(\sigma1-\sigma2)} - \frac{B2\ \text{EllipticPi}[\alpha2,\beta^2]}{D2 - f1(\sigma1-\sigma2)}\right) - \left(SeffdL + (\Delta1-\Delta2)\,\sigma1\,\sigma2 + L^2(\sigma1+\sigma2)\right)\frac{\Delta\lambda1}{L}\right);$$

$\Delta\lambda_{S_1{}^2}$

In[ ]:=
$$B1S1 = \frac{1}{2}\left(-L^2 S1\,\sigma1 + J\,S1^2\,\sigma1 - S1^3\,\sigma1 - S1\,\Delta2\,\sigma1^2 + J^2 S1\,\sigma2 - 2J\,S1^2\,\sigma2 + S1^3\,\sigma2 - J\,\Delta1\,\sigma1\,\sigma2 + 2\,S1\,\Delta1\,\sigma1\,\sigma2 + J\,\Delta1\,\sigma2^2 - S1\,\Delta1\,\sigma2^2\right);$$
$$B2S1 = \frac{1}{2}\left(L^2 S1\,\sigma1 + J\,S1^2\,\sigma1 + S1^3\,\sigma1 + S1\,\Delta2\,\sigma1^2 - J^2 S1\,\sigma2 - 2J\,S1^2\,\sigma2 - S1^3\,\sigma2 - J\,\Delta1\,\sigma1\,\sigma2 - 2\,S1\,\Delta1\,\sigma1\,\sigma2 + J\,\Delta1\,\sigma2^2 + S1\,\Delta1\,\sigma2^2\right);$$
$$D1S1 = -J\,S1 + S1^2 - \Delta1\,\sigma2;$$
$$D2S1 = \left(J\,S1 + S1^2 - \Delta1\,\sigma2\right);$$

$$\alpha1S1 = \frac{(-\sigma1)(f2-f1)}{D1S1 + f1\,\sigma1};$$
$$\alpha2S1 = \frac{(-\sigma1)(f2-f1)}{D2S1 + f1\,\sigma1};$$
$$\Delta\lambda4 = -\frac{1}{2S1}\left(\left(\frac{4}{\sqrt{A(f3-f1)}}\right)\left(-\frac{B1S1\ \text{EllipticPi}[\alpha1S1,\beta^2]}{D1S1 + f1\,\sigma1} + \frac{B2S1\ \text{EllipticPi}[\alpha2S1,\beta^2]}{D2S1 + f1\,\sigma1}\right) + S1(\sigma2-\sigma1)\,\Delta\lambda1\right);$$

$\Delta\lambda_{S_2{}^2}$

In[ ]:=
$$B1S2 = \frac{1}{2}\left(S2\,\sigma2\left(L^2 - J\,S2 + S2^2 - \Delta1\,\sigma2\right) - (J - S2)^2\,S2\,\sigma1 - (J - 2\,S2)\,\Delta2\,\sigma1\,\sigma2 + (J - S2)\,\Delta2\,\sigma1^2\right);$$

$$B2S2 = \frac{1}{2}\left(S2\,\sigma2\left(-L^2 - J\,S2 - S2^2 + \Delta1\,\sigma2\right) + (J + S2)^2\,S2\,\sigma1 - (J + 2\,S2)\,\Delta2\,\sigma1\,\sigma2 + (J + S2)\,\Delta2\,\sigma1^2\right);$$

$$D1S2 = (J - S2)\,S2 - \Delta2\,\sigma1;$$

$$D2S2 = -(J + S2)\,S2 - \Delta2\,\sigma1;$$

$$\alpha1S2 = \frac{(-\sigma2)\,(f2 - f1)}{D1S2 + f1\,\sigma2};$$

$$\alpha2S2 = \frac{(-\sigma2)\,(f2 - f1)}{D2S2 + f1\,\sigma2};$$

$$\Delta\lambda5 = -\frac{1}{2\,S2}\left(\left(\frac{4}{\sqrt{A\,(f3 - f1)}}\right)\left(-\frac{B1S2\,\text{EllipticPi}\left[\alpha1S2,\,\beta^2\right]}{D1S2 + f1\,\sigma2} + \frac{B2S2\,\text{EllipticPi}\left[\alpha2S2,\,\beta^2\right]}{D2S2 + f1\,\sigma2}\right) + S2\,(\sigma1 - \sigma2)\,\Delta\lambda1\right);$$

Fifth action J_5

In[ ]:=
$$I5 = \frac{1}{\pi}\left(SeffdL\,\Delta\lambda1 + J^2\,\Delta\lambda2 + L^2\,\Delta\lambda3 + S1^2\,\Delta\lambda4 + S2^2\,\Delta\lambda5\right);$$

In[ ]:=
```
dI5 = D[I5, SeffdL];
```
deriva

In[ ]:=
```
(*(*(*(*(*MinConst=NMinimize[{Abs[1/dI5],J>0,SeffdL>0},{SeffdL,J}]*)*)*)*)*)
```
minimiza apr··· valor absoluto

Spherical J_5

Assumming $\vec{L}$ along the z axis, thus $\theta_L = \phi_L = 0$

In[ ]:=
```
(*SeffdL=σ1 S1dL+σ2 S2dL*)
```

In[∘]:=
```
S1dL = S1 L (Cos[θ1] Cos[0] + Sin[θ1] Sin[0] Cos[ϕ1 - 0])
             coseno  coseno   seno    seno   coseno
S2dL = S2 L (Cos[θ2] Cos[0] + Sin[θ2] Sin[0] Cos[ϕ2 - 0])
             coseno  coseno   seno    seno   coseno
S1dS2 = S1 S2 (Cos[θ1] Cos[θ2] + Sin[θ1] Sin[θ2] Cos[Δϕ])
               coseno  coseno   seno    seno   coseno
```

Out[∘]=

$$\frac{21\,\mathrm{Cos}[\theta 1]}{100}$$

Out[∘]=

$$\frac{7\,\mathrm{Cos}[\theta 2]}{25}$$

Out[∘]=

$$\frac{3}{25}\,(\mathrm{Cos}[\theta 1]\,\mathrm{Cos}[\theta 2] + \mathrm{Cos}[\Delta\phi]\,\mathrm{Sin}[\theta 1]\,\mathrm{Sin}[\theta 2])$$

In[∘]:=
```
dI5sph =
  dI5 /. {SeffdL → (σ1 S1dL + σ2 S2dL)} /. {J → Sqrt[L^2 + S1^2 + S2^2 + 2 (S1dL + S2dL + S1dS2)]};
```

In[∘]:=
```
MinAngles = NMinimize[{Abs[1/dI5sph], 1 ≤ θ1 ≤ π, 0 ≤ θ2 ≤ π,
   minimiza apr···  valor absoluto
   1 / 100 ≤ Δϕ ≤ 49 / 100 × 2 π (*,(σ1 S1dL+σ2 S2dL)>0*)}, {θ1, θ2, Δϕ},
  WorkingPrecision → 50, PrecisionGoal → 30, MaxIterations → 500]
  precisión operativa   objetivo de precisión   máximo de iteraciones
```

Power: Infinite expression $\frac{1}{0 \times 10^{9}}$ encountered.

Power: Infinite expression $\frac{1}{0 \times 10^{9}}$ encountered.

NMinimize: The function value ComplexInfinity is not a number at {Δϕ, θ1, θ2} = {1.3840894758986420432601397072982411531213462531716, 2.1768038816709391492294685907657097722015014732988, 1.9947897704582908364495162546813380218148718061240}.

Out[∘]=

```
{0.072844280965561776226661894195757440684246830642223,
 {θ1 → 2.1768038816711456229717749451682834695756828863855,
  θ2 → 1.9947897704582205055356138558726141797580798005328,
  Δϕ → 1.3840894758984623400823998182167020822490680664177}}
```

In[∘]:=
```
Jphys = Sqrt[L^2 + S1^2 + S2^2 + 2 (S1dL + S2dL + S1dS2)] //. MinAngles[[2]]
```

Out[∘]=

0.600000000000000171542987939863054597792092908785 3

In[∘]:=
```
SeffdLphys = (σ1 S1dL + σ2 S2dL) //. MinAngles[[2]]
```

Out[∘]=

-0.525000000000000010305943246336020422513737630196 2

In[∘]:=
```
Re[1 / dI5 /. J → Jphys] /. SeffdL → SeffdLphys
```
parte real
```
Im[1 / dI5 /. J → Jphys] /. SeffdL → SeffdLphys
```
parte imaginaria

Out[∘]=

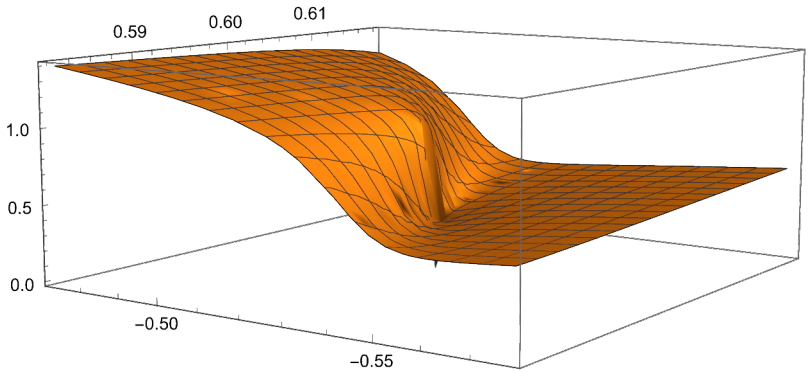
$0. \times 10^{-2}$

Out[∘]=

0

In[∘]:=
```
plot3D = Plot3D[Abs[1/dI5], {J, 0.97 Jphys, 1.03 Jphys}, {SeffdL, 0.9 SeffdLphys, 1.1 SeffdLphys}]
```
repres··· valor absoluto

Out[∘]=



In[∘]:=
```
MinAngles[[2]]
```

Out[∘]=

{θ1 → 2.1768038816711456229717749451682834695756828863855,
θ2 → 1.9947897704582205055356138558726141797580798005328,
Δϕ → 1.3840894758984623400823998182167020822490680664177}

In[•]:=
```
MinAnglesPert = {θ1 → 1.0012 ×
    2.16332713040107002981043886651723877802206946379291579050600483745670115250993`
     50., θ2 → 1.003 ×
    1.99935551705949533283774119207914094291486362758271432860879472870862789835976`
     50., Δϕ → 0.99 ×
    1.39525952005427776183986992373010611566727114508673385005522725040676978201378`50
JphysPert = Sqrt[L^2 + S1^2 + S2^2 + 2 (S1dL + S2dL + S1dS2)] //. MinAnglesPert
SeffdLphysPert = (σ1 S1dL + σ2 S2dL)    //. MinAnglesPert

verticalLine1 = Graphics3D[
                gráfico 3D
   {Red, Thickness[0.007], Line[{{Jphys, SeffdLphys, -6}, {Jphys, SeffdLphys, 6}}] }];
    rojo  grosor                línea
verticalLine2 = Graphics3D[{Blue, Thickness[0.007],
                gráfico 3D      azul  grosor
    Line[{{JphysPert, SeffdLphysPert, -6}, {Jphys, SeffdLphysPert, 6}}] }];
    línea
Show[plot3D, verticalLine1, verticalLine2]
muestra
```

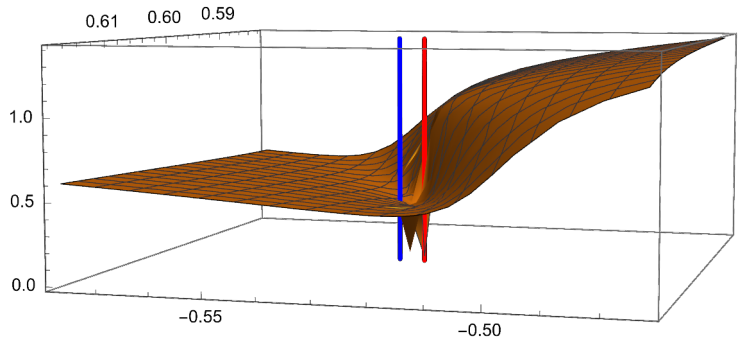
Out[•]=

0.599481

Out[•]=

-0.530241

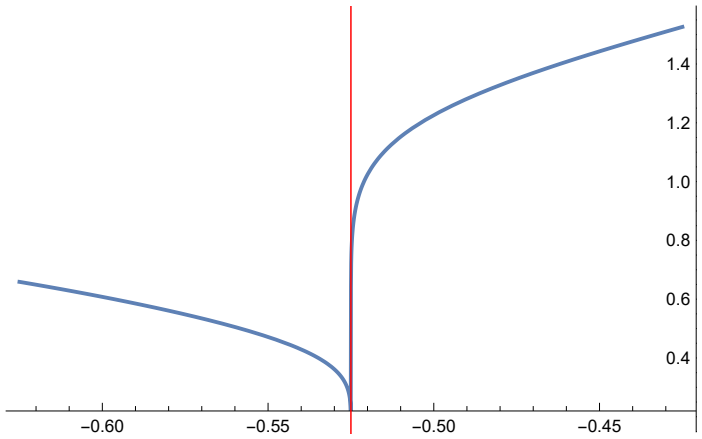
Out[•]=



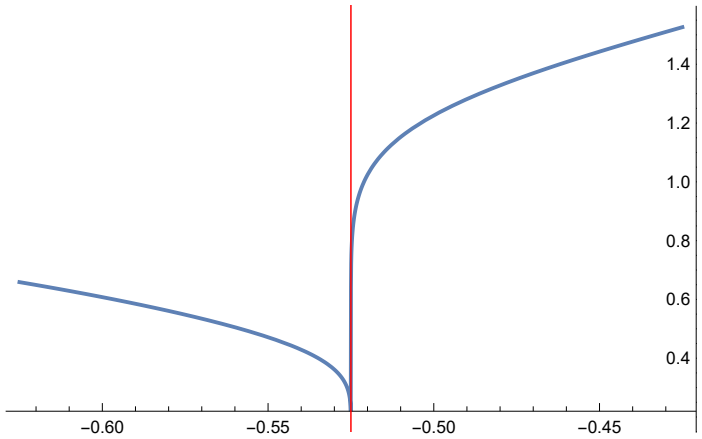
In[•]:=

In[•]:=
```
lowLim = SeffdLphys - 0.1; upLim = SeffdLphys + 0.1;
Plot[Abs[1 / dI5] /. J → Jphys, {SeffdL, lowLim, upLim},
repr··· valor absoluto
 Epilog → {Red, Line[{{SeffdLphys, -10^6}, {SeffdLphys, 10^6}}]}]
 epílogo  rojo  línea
Plot[Re[1 / dI5] /. J → Jphys, {SeffdL, lowLim, upLim},
repr··· parte real
 Epilog → {Red, Line[{{SeffdLphys, -10^6}, {SeffdLphys, 10^6}}]}]
 epílogo  rojo  línea
Plot[Im[1 / dI5] /. J → Jphys, {SeffdL, lowLim, upLim},
repr··· parte imaginaria
 Epilog → {Red, Line[{{SeffdLphys, -10^6}, {SeffdLphys, 10^6}}]}]
 epílogo  rojo  línea
```

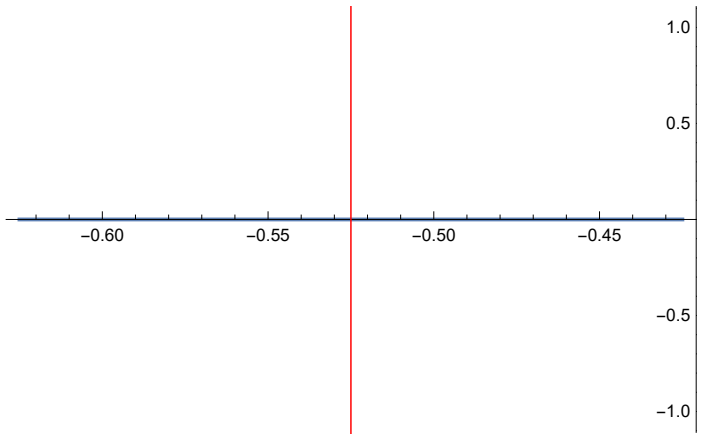
Out[ ]=


1.4
1.2
1.0
0.8
0.6
0.4
−0.60
−0.55
−0.50
−0.45

Out[ ]=

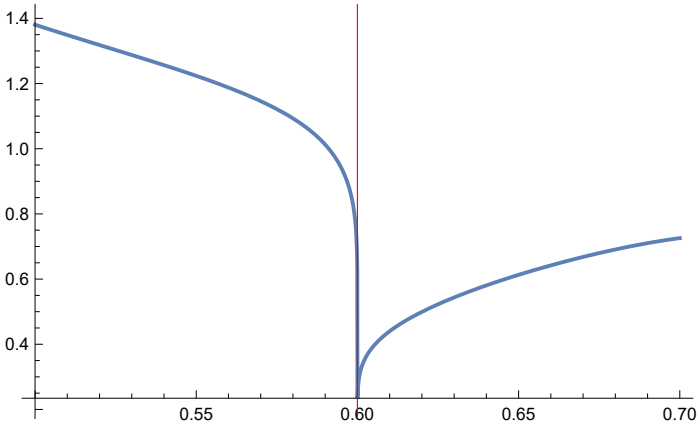

1.4
1.2
1.0
0.8
0.6
0.4
−0.60
−0.55
−0.50
−0.45

Out[ ]=

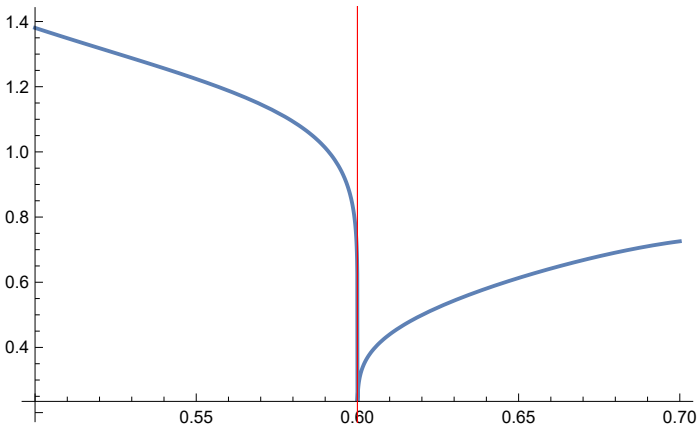

1.0
0.5
−0.60
−0.55
−0.50
−0.45
−0.5
−1.0

In[ ]:=
```
lowLim = Jphys - 0.1; upLim = Jphys + 0.1;
Plot[Abs[1 / dI5] /. SeffdL → SeffdLphys, {J, lowLim, upLim},
repr··· valor absoluto
 Epilog → {Red, Line[{{Jphys, -10^6}, {Jphys, 10^6}}]}]
 epílogo  rojo  línea
Plot[Re[1 / dI5] /. SeffdL → SeffdLphys, {J, lowLim, upLim},
repr··· parte real
 Epilog → {Red, Line[{{Jphys, -10^6}, {Jphys, 10^6}}]}]
 epílogo  rojo  línea
Plot[Im[1 / dI5] /. SeffdL → SeffdLphys, {J, lowLim, upLim},
repr··· parte imaginaria
 Epilog → {Red, Line[{{Jphys, -10^6}, {Jphys, 10^6}}]}]
 epílogo  rojo  línea
```

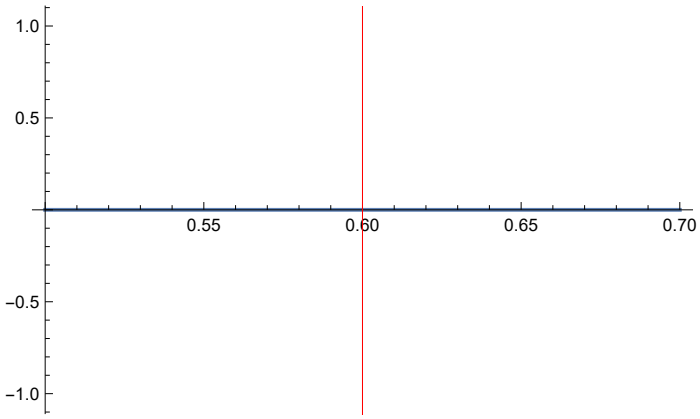
Out[ ]=



Out[ ]=



Out[•]=



In[•]:=
```
Roots1 = roots /. J → Jphys /. SeffdL → SeffdLphys
Roots2 = roots /. J → JphysPert /. SeffdL → SeffdLphysPert
```

Out[•]=

$\{f_1 \to -0.05333333333333333831975376447324952144796616987 52,$
$f_2 \to 0.07999999999999998917957827172540 42,\ f_3 \to 0.08000000000000000139634024113639 77\}$

Out[•]=

$\{f_1 \to -0.0541179,\ f_2 \to 0.0776967,\ f_3 \to 0.080009\}$

•